
Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

Anonymous Author(s)

Affiliation

Address

email

Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, tying the learned function to one sampling lattice and leaving expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance.

Across 22 sections from four technologies (336 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, neural-field and imputation baselines. Because twelve of those sections come from one brain, we take the *specimen* as the unit of analysis and assemble 10 independent ones: the advantage survives Holm correction across the baseline family against 3 of them, including a graph model. Two controls localize it. An exactly parameter-matched ablation, in which the 0.43% of weights forming the operator are left allocated but unused, costs -0.023 Pearson r ; and a non-spatial baseline sharing the read-out head shows message passing contributes at most 0.002 against 0.014 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, delimiting what any method of this class can identify from static tissue.

Two limits are worth stating plainly. The separation is unresolved against the strongest graph baseline (STAGATE-style: 7/10 specimens, $p = 0.43$), so the method is distinguishable from most of this literature but not all of it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a limitation a spectral-matching term bounds but does not remove.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on

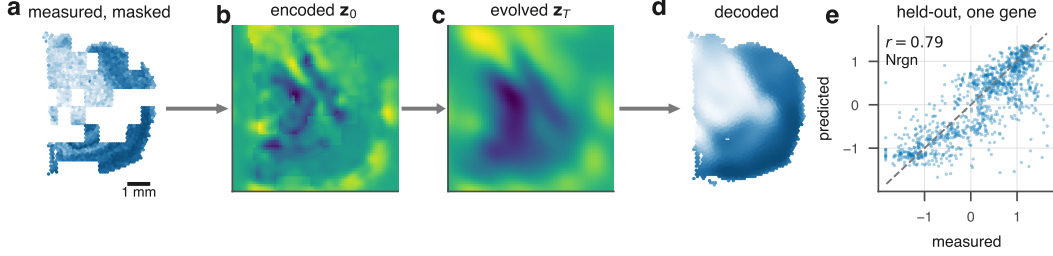


Figure 1: **Neural Reaction-Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction-diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction-diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction-Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction-diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction-diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions ??–1); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts our perturbation result.

2 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode-evolve-decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are

splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral blocks [Li et al., 2021] refine the field.

The reaction–diffusion operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output.

Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-step map that is second-order accurate (Proposition ??). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition ??). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters, and no projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component (Proposition ??). Both are proved in Appendix ?? and checked numerically in Table ??; Section ?? measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 1 makes this precise: the residual along the model’s own trajectory is $\frac{\Delta t}{2} \|\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta\| \mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix A). We instead require that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_\mathbf{k} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_\mathbf{k}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Re spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix ??).

3 Experimental setup

Data. We use 12 public datasets spanning four spatial technologies (Table 1), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section ?? . Raw reads are never used:

we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force- includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Table 1: **Benchmark composition.** One row per assay. *Specimens* is the number of independent biological samples, which is the unit the conservative analysis uses and the quantity that limits its resolution: the MERFISH sections are serial sections of a single brain and count once. Sampling density spans two orders of magnitude, from multi-cell spots to single cells, which is the range over which a continuous formulation is supposed to be an advantage. Per-section inventory in Appendix ??.

assay	organism	specimens	sections	locations	genes	resolution
MERFISH	Mus	1	12	900,614	500–503	single cell
Xenium In Situ	Mus	1	1	36,362	248	single cell
Stereo-seq	Mus	1	2	24,258	2,069–2,099	bin50 (25 um)
Visium	Homo/Mus	5	5	15,914	2,073–2,087	spot (multi-cell)
Visium FFPE	Homo/Mus	2	2	6,633	2,075–2,088	spot (multi-cell)
total		10	22	983,781		

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head — the most favorable choice available — and is marked *-style*, since these are not reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and carries every claim about generality; the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes, kept distinct. Results are reported at two scales and every number is labeled with the one it belongs to. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and is the basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

4 Results

4.1 Masked spatial reconstruction across 22 sections

Table 2: **Masked spatial reconstruction across 22 sections and 10 independent specimens.** Left: mean $\pm$ s.d. over sections, with Holm-corrected stars from the paired section-level test. Right: the conservative analysis, in which serial sections of one specimen are collapsed before testing — specimens won, Cohen’s d_z , and the Holm-corrected p over the baseline family, bold where $p < 0.05$. $|\Delta I|$ is the error in Moran’s I , on which NRDO is worst: the predicted fields are too smooth. Section-level $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

model	Pearson $r \uparrow$	$ \Delta I \downarrow$	specimens	d_z	p_{Holm}
STAGATE-style	$0.170 \pm 0.064^{**}$	0.769 ± 0.045	7/10	0.45	0.43
Autoencoder (non-spatial)	$0.169 \pm 0.063^{***}$	0.761 ± 0.049	7/10	0.65	0.21
GNN (GCN)	$0.167 \pm 0.061^{***}$	0.793 ± 0.046	9/10	1.05	0.020
Tangram-style	$0.163 \pm 0.064^{***}$	0.805 ± 0.054	9/10	0.92	0.08
GP, multi-bandwidth	$0.133 \pm 0.053^{***}$	0.770 ± 0.040	10/10	2.12	0.014
Neural field (SIREN)	$0.118 \pm 0.051^{***}$	0.612 ± 0.064	10/10	1.93	0.014
NRDO (ours)	0.183 ± 0.063	0.848 ± 0.047	—	—	—

NRDO improves masked-region reconstruction over every baseline tested, on 22 sections spanning four technologies (336 runs), and the improvement survives the conservative unit of analysis (Table 2).

Two units are reported because they answer different questions. With the *section* as the unit every model sees identical held-out blocks, so a paired test removes the between-section variance that otherwise dominates; all 7 baselines are beaten with Holm-corrected $p \leq 0.003$. But twelve of these sections are serial sections of one brain, and treating them as independent overstates the effective sample size. Collapsing to 10 independent *specimens* costs most of that power and is the defensible test: the separation still survives Holm–Bonferroni correction across the family of 7 baselines against 3 of them (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN)), winning on at least 9/10 specimens at $p \leq 0.020$. The tightest, GNN (GCN), has a bootstrap 95% CI of $[+0.0051, +0.0166]$ excluding zero. It does not survive against STAGATE-style (Section 6). The grouping is conservative and does not drive the result: counting the two Stereo-seq stages separately gives 11 specimens and the same 3 significant comparisons.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.169 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to 0.002 Pearson r , against 0.014 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms (Section 5).

Spatial autocorrelation. Predicted maps beside measured ones are in Appendix ??, Fig. ?. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.335 against 0.347), and it has by some margin the largest error in Moran’s I (0.848 against 0.612 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

175 **Further results.** The exponential integrator needs $32\times$ fewer steps than explicit Euler for the same
 176 tolerance; the reconstruction does not degrade spatial-domain structure; zero-shot transfer to a new
 177 tissue does not exceed the training-mean floor; and a counterfactual perturbation probe returns the
 178 null result Theorem 2 predicts. All are reported in full in the archival version.

179 5 Ablations

180 The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is
 181 never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (-0.023), and
 182 removing the reaction gives 0.241 ± 0.003 (-0.006): disabling the operator costs more than the
 183 entire margin over the best baseline. The control is exactly parameter-matched — the operator holds
 184 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused
 185 — so architecture, parameter count and optimizer budget are identical and the difference cannot
 186 be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the
 187 control separates the two explanations. The full sweep, including the biological regularizers and the
 188 PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in
 189 Appendix ??.

190 6 Limitations

191 **Identifiability.** A single stationary snapshot cannot identify a dynamical system. Theorem 2 makes
 192 this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a
 193 Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather
 194 than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than
 195 physical morphogen transport; Section 4 shows they are moreover close to their initialization, which
 196 is why we make no claim about signaling. Dense temporal or interventional sampling would remove
 197 this degeneracy.

198 **Over-smoothing.** The predicted fields are smoother than the measurements, and on Moran’s I error
 199 NRDO trails the baselines it leads on the other metrics. This survives training to convergence and
 200 is only partly correctable by a spectral-matching term (Section ??), so it bounds the applications
 201 we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp
 202 boundaries less so.

203 **One baseline remains unresolved.** The benchmark’s statistical resolution is set by independent
 204 specimens, not sections: twelve of 22 sections are serial sections of one brain. An earlier version of
 205 this work reported five specimens, at which no family-wise significant result was reachable at any
 206 effect size. We therefore added tissues rather than sections, reaching 10, and the separation now
 207 survives Holm–Bonferroni correction across the family of 7 baselines against 3 of them including
 208 GNN (GCN) ($p = 0.020$, 9/10 specimens). It does not against STAGATE-style (7/10, $p = 0.43$),
 209 whose attention-weighted neighbourhood aggregation is the closest existing analogue to what the
 210 operator does. We take that as the honest boundary of the claim: the continuous formulation is
 211 distinguishable from most of this literature and not from all of it. One section (visium_human_heart)
 212 is excluded from every specimen-level analysis because its held-out set falls below 200 locations, at
 213 which Pearson r cannot be estimated to the precision the comparison requires.

214 **Transfer.** Under the held-out-block protocol the fitted operator does not transfer to a new tissue at
 215 all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline
 216 transfers better. It is a within-section model, and we no longer claim otherwise.

217 **Scope.** The *-style* baselines implement each method’s architectural core with a common read-out
 218 head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget
 219 and subsampled sections so that 336 runs were feasible on CPU, and scope is limited to single 2-D
 220 sections.

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286 A Vacuity of the along-trajectory residual

287 Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own
 288 trajectory. The following makes precise why that penalty is uninformative for the present architec-
 289 ture, and motivates the steady-state formulation adopted instead.

290 **Proposition 1** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$
 291 $\Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-
 292 ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (4)$$

293 where $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,
 294 and $R_{\Delta t}$ does not depend on the observed data.

295 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
 296 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
 297 evaluated at the same state $\mathbf{z}$. $\square$

298 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
 299 observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts
 300 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
 301 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
 302 scaling is confirmed in Table ?? (observed order 1.00).

303 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
 304 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
 305 constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space,
 306 characterized next.

307 B Non-identifiability from a single snapshot

308 The following records the central epistemic limitation of the setting: an operator fitted to one station-
 309 ary field is determined only on a measure-zero subset of its own domain. The argument is elementary
 310 — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint
 311 simply says nothing off that image — and we state it as a theorem for its consequence rather than
 312 for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in
 313 this literature: it means every counterfactual an operator of this class produces is a statement about
 314 the architecture, not about the data, and it predicts in advance the null result of Section ??.

315 **Theorem 2** (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose*
 316 *$C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (5)$$

317 *If $\mathcal{F}(\mathbf{D}, \mathbf{z}^*) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for*
 318 *every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the*
 319 *space of such φ is infinite-dimensional.*

320 *Proof.* The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it
 321 whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$
 322 is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets
 323 of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$
 324 is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ;
 325 choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a
 326 linearly independent family, all vanishing on $\mathcal{R}$. $\square$

327 **Corollary 3** (What a stationarity fit can determine). *Under the hypotheses of Theorem 2, the steady-*
 328 *state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about*
 329 *f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the*
 330 *observed manifold—is determined by the inductive bias of the architecture and the regularizers, not*
 331 *by the data.*

332 Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section ?? dis-
 333 places $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states
 334 off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is
 335 therefore consistent with the theory rather than merely an empirical disappointment: a single sta-
 336 tionary snapshot does not contain the information such a counterfactual would require. Breaking the
 337 degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal
 338 sampling, or interventional data.